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University

NAAC A++
ACCREDITED UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY
BACHELOR OF TECHNOLOGY

OBJECT ORIENTED PROGRAMMING WITH JAVA LABORATORY
(03010803PC04)

III SEMESTER

Department of Information Technology



Laboratory Manual
Session 2026-27

CERTIFICATE

This is to certify that

Mr./Ms.....

with enrolment nohas successfully
completed his/her laboratory experiments in the **OOPJ-LAB (03010803PC04)**
from the department of **INFORMATION TECHNOLOGY** during the
academic year 2026-2027.



Date of Submission:.....

Staff In charge:.....

Head Of Department:.....



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Sr. No	Experiment Title	Page No		Date of Completion	Marks	Sign
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1.	Write a Java program to display “Hello World” and demonstrate the Java program structure and demonstrate use of variables, data types, and type casting.					
2.	Write a Java program to perform arithmetic, relational, and logical operations, bitwise and other operators.					
3.	Write a Java program using below conditional statements. a) Even or Odd using if.. else statement. b) Roots of Quadratic Equation using else if ladder. c) Largest of three numbers using nested if else. d) Find out the week day using switch statement.					
4.	Write a Java program to demonstrate looping constructs: a) Reverse of a number using while loop. b) Prime number using do while loop. c) nth term of fibonacci sequence using for loop.					
5.	Write a Java program to perform methods and example on recursion.					
6.	Write a Java program to perform operations on one-dimensional and multi-dimensional arrays.					
7.	Write a Java program to demonstrate 1D - Arrays & 2-D Arrays. a) Maximum value and Second Maximum value without duplicates. b) Sort the names in Ascending Order. c) Addition of two matrix. d) 3x3 Matrix Multiplication.					
8.	Write a Java program to demonstrate string handling using String, StringBuffer, and StringBuilder.					
9.	Write a Java program to check the word is palindrome or not.					



10.	Write a Java program to create a class and object and demonstrate constructors.					
11.	Write a Java program to demonstrate encapsulation using access specifiers.					
12.	Write a Java program to demonstrate inheritance: a) Single Inheritance. b) Multilevel Inheritance. c) Hierarchical Inheritance. d) Hybrid Inheritance.					
13.	Write a Java program to demonstrate use of the this and super keywords					
14.	Write a Java program to demonstrate polymorphism (method overloading and overriding).					
15.	Write a Java program to implement abstraction using abstract classes and interfaces(Multiple Inheritance).					
16.	Write a Java program to demonstrate exception handling using try-catch-finally blocks.					
17.	Write a Java program to demonstrate exception handling using try-catch-finally blocks.					
18.	Write a Java program to implement List interface using ArrayList and LinkedList.					
19.	Write a Java program to implement Set interface using HashSet and TreeSet.					
20.	Write a Java program to implement Queue interface using Priority Queue and Deque.					
21.	Write a Java program to implement map interface using HashMap and LinkedHashMap.					
22.	Write a Java program to demonstrate multithreading using Thread class and Runnable interface.					